

Section 5.5 Similarity and Diagonalization

Exercise 5.5.1

By computing the trace, determinant, and rank, show that A and B are not similar in each case.

d. $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$

e. $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ -3 & 6 & -3 \end{bmatrix}$

Exercise 5.5.4

In each case, decide whether the matrix A is diagonalizable. If so, find P such that $P^{-1}AP$ is diagonal.

c. $A = \begin{bmatrix} 3 & 1 & 6 \\ 2 & 1 & 0 \\ -1 & 0 & -3 \end{bmatrix}$

Exercise 5.5.7

Let λ be an eigenvalue of A with corresponding eigenvector x . If $B = P^{-1}AP$ is similar to A , show that $P^{-1}x$ is an eigenvector of B corresponding to λ .

Exercise 5.5.10

Let A be a diagonalizable $n \times n$ matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (including multiplicities). Show that:

a. $\det A = \lambda_1 \lambda_2 \dots \lambda_n$

b. $\operatorname{tr} A = \lambda_1 + \lambda_2 + \dots + \lambda_n$

Section 6.1 Examples and Basic Properties

Exercise 6.1.2

Are the following sets vector spaces with the indicated operations? If not, why not?

c. The set of all polynomials of degree ≤ 3 ; operations of P .

d. The set $\{1, x, x^2, \dots\}$; operations of P .

h. The set V of real numbers; usual operations.

Exercise 6.1.6

In each case show that the condition $au + bv + cw = 0$ in V implies that $a = b = c = 0$.

b. $V = M_{22}; \quad u = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad v = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad w = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

c. $V = P; \quad u = x^3 + x, \quad v = x^2 + 1, \quad w = x^3 - x^2 + x + 1$

Exercise 6.1.9

Show that $-0 = 0$ in any vector space. Cite all axioms used.

Exercise 6.1.10

Show that the zero vector 0 is uniquely determined by the property in axiom A4.

Exercise 6.1.11

Given a vector v , show that its negative $-v$ is uniquely determined by the property in axiom A5.